AQ6.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Suppose $T: \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which of the following options are correct?

- [ ] The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1,0),(0,1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- [ ] The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1,0),(0,1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.
- [ ] T is neither one-one nor onto.
- [ ] T is one-one but not onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The matrix corresponding to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1,0),(0,1)\}$, for both the domain and co-domain is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

T is neither one-one nor onto.

1 point
If the matrix corresponding to a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$, with respect to standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1,0),(0,1)\}$, for both the domain and co-domain, is $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$. Which of the following is the appropriate definition of T?

- ( ) $T(x, y) = (2x + y, 3x + 4y)$
- ( ) $T(x, y) = (2x + y, 3x - 4y)$
- ( ) $T(x, y) = (x + 4y, 2x + 3y)$
- ( ) $T(x, y) = (2x + 3y, x + 4y)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y) = (2x + 3y, x + 4y)$

Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of $\mathbb{R}^3$. Let $\beta = \{(2,1,0),(1,0,1)\}$ be a basis of W. Let $T: W \to \mathbb{R}^2$ be a linear transformation such that $T(2,1,0) = (1,0)$ and $T(1,0,1) = (0,1)$. Answer the questions 3, 4 and 5 using the given information.

1 point

Which of the following is the appropriate definition of T?

☐ $T(x, y, z) = (x, y)$

☐ $T(x, y, z) = (x, z)$

☐ $T(x, y, z) = (y, z)$

☐ $T(x, y, z) = (x - y - z, x - 2y)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y, z) = (y, z)$
$T(x, y, z) = (x - y - z, x - 2y)$

1 point

Choose the correct options.

○ T is one to one but not onto.

○ T is onto but not one to one.

○ T is neither one to one nor onto.

○ T is an isomorphism.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is an isomorphism.

1 point

What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for R^2?

○ $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

○ $\frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

1 point

Choose the set of correct options.

☐ Let $u = (3, 1, 0)$, $v = (0, 1, 7)$, and $w = (3, 0, -7)$. There is a linear transformation $T: R^3 \to R^3$ such that $T(u) = T(v) = (0,0,0)$ and that $T(w) = (5,1,0)$.

☐

Let $u = (2,1,0)$, $v = (1,0,1)$, and $w = (3,5,6)$. There is a linear transformation $T: R^3 \to R^2$ such that $T(u) = (1,0)$, $T(v) = (0,1)$ and that $T(w) = (5,6)$.

☐

Let $u = (1,0,0)$, $v = (1,0,1)$, and $w = (0,1,0)$. There is a linear transformation $T: R^3 \to R^3$ such that $T(u) = (0,0,0)$, $T(v) = (0,0,0)$, $T(w) = (0,0,0)$, and $T(1,4,2) = (2,4,1)$.

☐

Let $u = (1,0)$, and $v = (0,1)$,. There is a linear transformation $T: R^2 \to R^2$ such that $T(u) = (\pi, 1)$, and $T(v) = (1, e)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Let $u = (2,1,0)$, $v = (1,0,1)$, and $w = (3,5,6)$. There is a linear transformation $T: R^3 \to R^2$ such that $T(u) = (1,0)$, $T(v) = (0,1)$ and that $T(w) = (5,6)$.

Let $u = (1,0)$, and $v = (0,1)$,. There is a linear transformation $T: R^2 \to R^2$ such that $T(u) = (\pi, 1)$, and $T(v) = (1, e)$.

Level 2

1 point

Choose the set of correct options.

☐

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \to R^2$, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of T with respect to any other basis β for both the domain and co-domain.

☐

Let T be an isomorphism between two vector spaces, and A be the matrix representation of T with respect to some basis. Then the nullity of A is 0.

☐

If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T: R^2 \to R^2$ with respect to the standard ordered basis of R^2, for both the domain and co-domain, then the matrix representation of $T \circ T$ with respect to the same bases will also be A.

☐

If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T: R^2 \to R^2$ with respect to the standard ordered basis of R^2, for both the domain and co-domain, then the matrix representation of $T \circ T$ will be the identity matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \to R^2$, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of T with respect to any other basis β for both the domain and co-domain.

Let T be an isomorphism between two vector spaces, and A be the matrix representation of T

with respect to some basis. Then the nullity of A is 0.
If $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the matrix representation of $T: \mathbb{R}^2 \to \mathbb{R}^2$ with respect to the standard ordered basis of $\mathbb{R}^2$, for both the domain and co-domain, then the matrix representation of $T \circ T$ with respect to the same bases will also be A.

Let $\beta = \{(1,0),(0,1)\}$ and $\gamma = \{(1,1),(1,-1)\}$ be two bases of $\mathbb{R}^2$. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ with respect to β for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

1 point

What is the matrix representation of T with respect to γ for the domain and β for the co-domain?

- ○ $\begin{bmatrix} a+b & c+d \\ a-b & c-d \end{bmatrix}$
- ○ $\begin{bmatrix} a-b \\ c-d \end{bmatrix}$
- ○ $\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$
- ○ $\begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

1 point

What is the matrix representation of T with respect to γ for both the domain and the co-domain?

- ○ $\begin{bmatrix} a+b+c+d & a-b+c-d \\ a+b-c-d & a-b-c+d \end{bmatrix}$
- ○ $\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a-b+c-d}{2} \\ \frac{a+b-c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$
- ○ $\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a+b-c-d}{2} \\ \frac{a-b+c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$
- ○ $\begin{bmatrix} a+b+c+d & a+b-c-d \\ a-b+c-d & a-b-c+d \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix} \frac{a+b+c+d}{2} & \frac{a-b+c-d}{2} \\ \frac{a+b-c-d}{2} & \frac{a-b-c+d}{2} \end{bmatrix}$

1 point

Which of the following options are correct?

- ☐ If $T: \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation which is both one-one and onto then T^{-1}

is also a linear transformation.

☐

If $T: R^n \to R^m$ then the matrix corresponding to T, with respect to the standard bases for R^n and R^m is of order $n \times m$.

☐

If $T: R^n \to R^m$ then the matrix corresponding to T, with respect to the standard bases for R^n and R^m is of order $m \times n$.

☐

Consider the vector space $V = \{\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in R, \}$, that is, the set of all 2×2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from R^3 to V.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $T: R^2 \to R^2$ is a linear transformation which is both one-one and onto then T^{-1} is also a linear transformation.

If $T: R^n \to R^m$ then the matrix corresponding to T, with respect to the standard bases for R^n and R^m is of order $m \times n$.

Consider the vector space $V = \{\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in R, \}$, that is, the set of all 2×2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from R^3 to V.

Check Answers

Your score is: 0/10